

Wrap Up

In this project we have found sources online to help us with code Linear Modelling, Random Forests and Gradient Boosting Regression on Python and successfully implement it in order to analyse a dataset about factors affecting educational outcomes.

In the first section we tried to produce code in Python that would allow us to fit a linear model to our data and assess how good this model was using a combination of online sources, as well as prior knowledge and generative AI. We were able to produce a function to carry out stepwise selection and fit linear models successfully. However, the plots suggested that using a linear model is not an effective way of modelling this data specifically, due to violations of the assumptions in a linear model.

In the second section we tried to use Random Forests to analyse the data. We were able to produce random forests which predicted final grades very accurately when previous academic performance was included, but performance dropped substantially after removing G1, G2 and previous failures. This suggests that previous academic performance contains most of the useful information for predicting G3. Using only the remaining variables, the random forest we were able to produce still performed better than the baseline model, but its predictive accuracy was fairly limited. Cross-validation also showed that performance varied between different subsets of students. Hyperparameter tuning improved the test R^2 slightly, but the large difference between training and test performance remained. Among the remaining predictors, intention to pursue higher education, school, study time and some behavioural variables were the most useful to the tuned model.

Finally, we tried to create a Gradient Boosting Model. This model achieved an R^2 value of approx. 0.155, meaning it explained only a small proportion of the variation in students' final grades. This suggests the variables included in the model have limited ability to predict G3 accurately on their own. This model achieved a Mean Absolute Error of approximately 2.15 and the Root Mean Squared Error was approximately 2.87. The model produced an R^2 value of 0.155, indicating that it explained approx. 15.5% of the variation in final grades within the test data. This suggests that after excluding G1, G2 and previous failures the remaining data contains useful information, but is not sufficient to predict students' final grades very accurately. The plots provide a representation of these prediction errors. This analysis used a 80/20 train-test split. The measured performance therefore depends partly on which students were randomly allocated to the training and testing sets. Using cross-validation in future work would provide a more reliable estimate. The Gradient Boosting parameters, such as the

number of trees and max tree depth, were also chosen in advance rather than being systematically optimised. The next step here could be to investigate the effect of changing parameter combinations on the quality of the predictions.

In summary, we were able to find information on and implement various data modelling methods using a range of sources, creating models that could be easily adapted to other datasets. However, the results we got for our chosen dataset suggest that previous academic performance may contain important information for predicting final grades that is not fully captured by the remaining student characteristics included in the data.